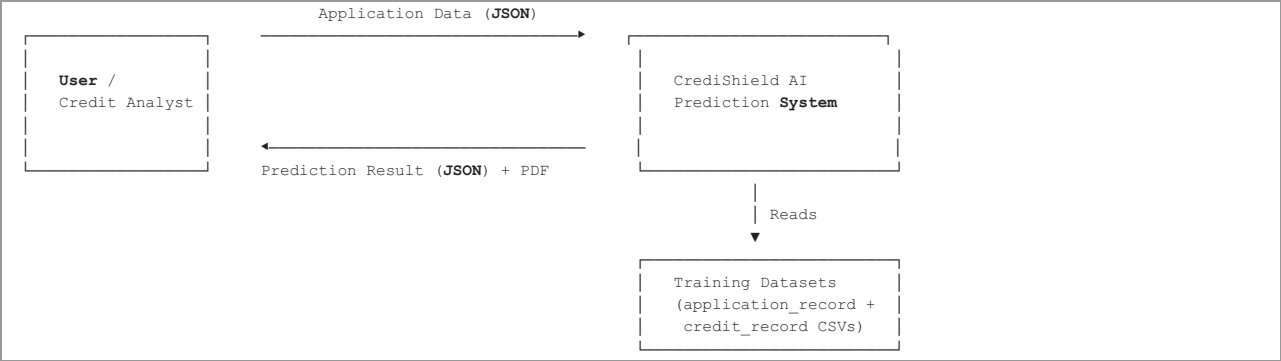


Data Flow Diagram

CrediShield AI — Credit Card Approval Prediction System

Level 0 — Context Diagram

The context diagram shows the system as a single process interacting with external entities.

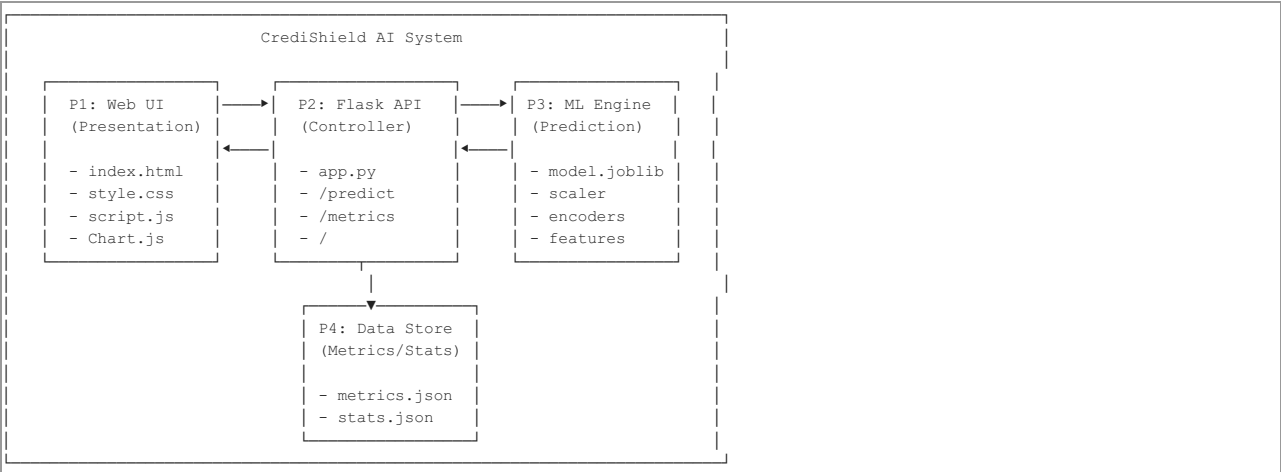


External Entities:

- User / Credit Analyst:** Submits applicant data and receives predictions
- Training Datasets:** Kaggle CSV files used during model training phase

Level 1 — System Decomposition

The system is decomposed into four major processes.

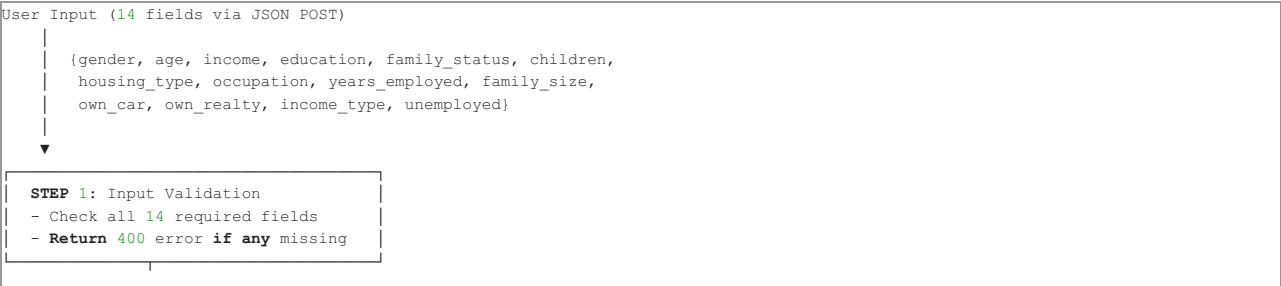


Process Descriptions:

Process	Name	Input	Output	Technology
P1	Web UI (Presentation Layer)	User interactions (clicks, form entries)	HTTP requests to Flask API	HTML5, CSS3, JavaScript, Chart.js
P2	Flask API (Controller Layer)	HTTP requests (JSON payloads)	Processed predictions, metrics data	Flask 3.0.3, Python 3.12
P3	ML Engine (Prediction Layer)	18-element feature vector	Prediction (0/1) + probability	Scikit-Learn, Gradient Boosting
P4	Data Store (Persistence Layer)	Training outputs	Metrics and statistics for charts	JSON files

Level 2 — Detailed Prediction Data Flow

This diagram shows the step-by-step data transformations inside the /predict endpoint.



| ☒ All fields present

STEP 2: Parse Numerical Values

```
- age → int
- income → float
- children → int
- family_size → int
- years_employed → float
- unemployed → int (0 or 1)
```

STEP 3: Feature Engineering

```
- income_per_member = income /
  family_size
- income_cat: Low/Medium/High
  (thresholds: $100K, $200K)
- age_cat: Young/Middle-Aged/Senior
  (thresholds: 30, 50)
- employed_cat: Unemployed/Junior/
  Mid-level/Senior
  (thresholds: 0, 5, 15 years)
```

STEP 4: Categorical Encoding

```
LabelEncoder.transform() for:
- CODE_GENDER (M/F → 0/1)
- FLAG_OWN_CAR (N/Y → 0/1)
- FLAG_OWN_REALTY (N/Y → 0/1)
- NAME_INCOME_TYPE (5 categories)
- NAME_EDUCATION_TYPE (5 cats)
- NAME_FAMILY_STATUS (5 cats)
- NAME_HOUSING_TYPE (6 cats)
- OCCUPATION_TYPE (19 cats)
- INCOME_CAT (3 categories)
- AGE_CAT (3 categories)
- EMPLOYED_CAT (4 categories)
Total: 11 encoded features
```

STEP 5: Numerical Scaling

```
StandardScaler.transform() for:
- CNT_CHILDREN
- AMT_INCOME_TOTAL
- CNT_FAM_MEMBERS
- AGE
- YEARS_EMPLOYED
- INCOME_PER_MEMBER
Total: 6 scaled features
```

STEP 6: Feature Vector Assembly

```
- Read feature_names.joblib
- Arrange all 18 features in the
  exact order used during training
- Create numpy array: shape (1, 18)
```

STEP 7: Model Inference

```
- model.predict(features)
  → 0 (Rejected) or 1 (Approved)
- model.predict_proba(features)
  → [prob_rejected, prob_approved]
```

STEP 8: Risk Classification

```
If Approved:
  confidence = prob_approved × 100
  ≥ 85% → Low Risk
  ≥ 65% → Medium Risk
  < 65% → High Risk
If Rejected:
  confidence = prob_rejected × 100
  risk_level = "High"
```



Data Store Descriptions

Data Store	File	Contents	Created By	Used By
DS1	card_model.joblib	Trained GradientBoostingClassifier (136 KB)	train.py	app.py (prediction)
DS2	scaler.joblib	Fitted StandardScaler object (1 KB)	train.py	app.py (numerical scaling)
DS3	label_encoders.joblib	Dictionary of 11 LabelEncoder objects (3.5 KB)	train.py	app.py (categorical encoding)
DS4	feature_names.joblib	Ordered list of 18 feature column names (0.3 KB)	train.py	app.py (vector assembly)
DS5	metrics.json	Model comparison scores for all 6 models (3.8 KB)	train.py	app.py → /metrics endpoint
DS6	stats.json	Training dataset statistics (1.1 KB)	train.py	app.py → /metrics endpoint
DS7	localStorage	Prediction history log (browser-side)	script.js	script.js (history tab)

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